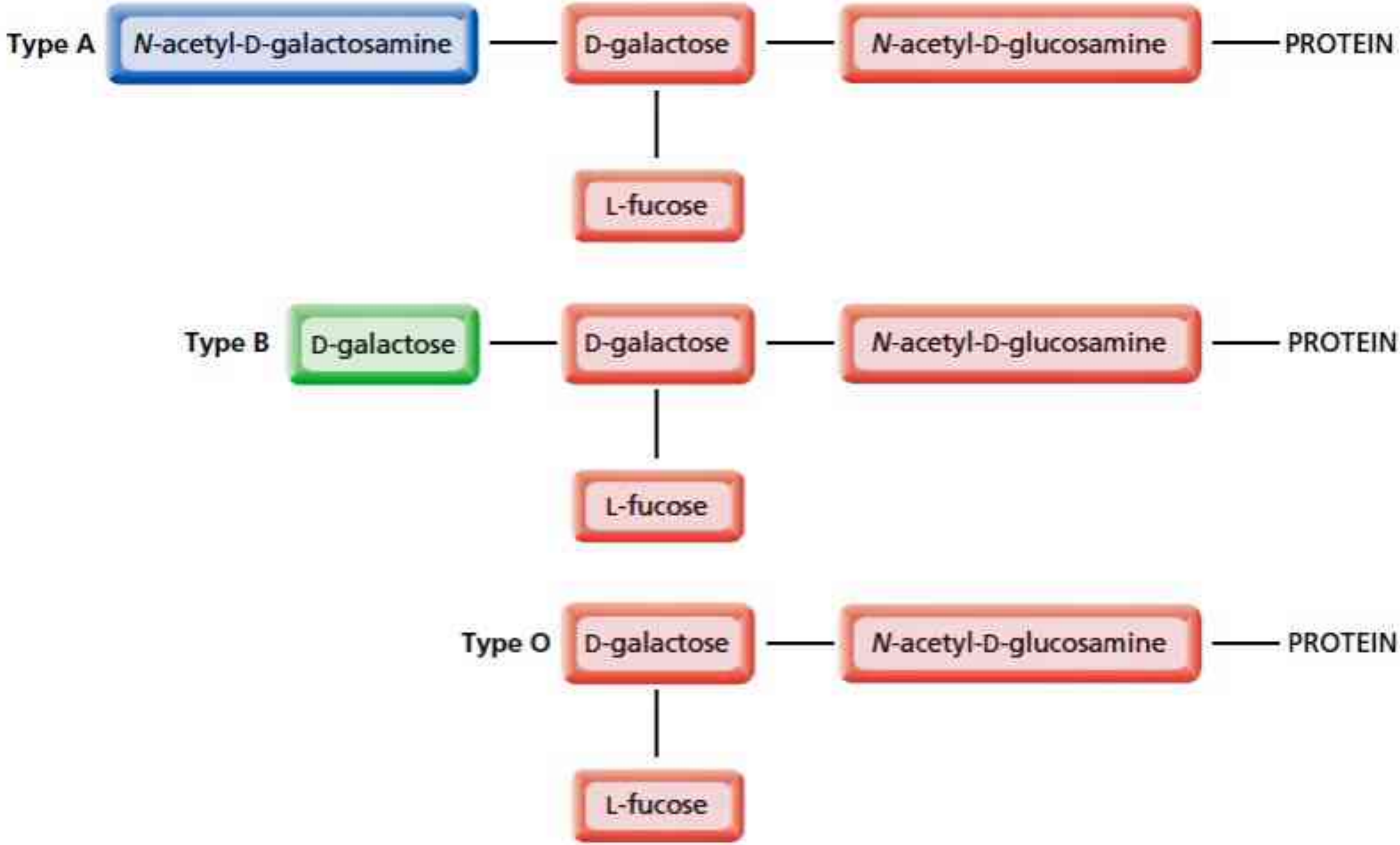
Heteropolysachrides

- Consist of two or more types of monomers.
- I-Glycos-amino-glycans(GAGs)/Mucopoly sacharides
- Hyaluronic acid,Chondroitin sulphsates,Heparin,Blood group poly.and,S.mucoids.
- II-Glyco-conjugates
- Proteoglycans,Glycoproteins and Glycolipids
- III-Mucilages
- Agar,Vegetable gums and Pectins

2- Glycoprotein

	Glycoproteins	Proteoglycans
Structure	Protein mainly associated with small amount of CHO	CHO mainly associated with small amount of protein
CHO chain	-Branched -Short chain(2-10 residues) -Don't have serial repeats -May or may not be negatively charged	-Linear -Long chain(200 disaccharide units) Have repeating -Negatively charged -disaccharide units



▲ Figure 22.5
Blood type is determined by the nature of the sugar on the surfaces of red blood cells. Fucose is 6-deoxygalactose.

Looking at Figure 22.5, we can see why the immune system of type A people recognizes type B blood as foreign and vice versa. The immune system of people with type A, B, or AB blood does not, however, recognize type O blood as foreign, because the carbohydrate in type O blood is also a component of types A, B, and AB blood. Thus, anyone can accept type O blood, so people with type O blood are called universal donors. Type AB people can accept types AB, A, B, and O blood, so people with type AB blood are referred to as universal acceptors.

PROBLEM 29◆

From the nature of the carbohydrate bound to red blood cells, answer the following questions:

a. People with type O blood can donate blood to anyone, but they cannot receive blood from everyone. From whom can they *not* receive blood?

b. People with type AB blood can receive blood from anyone, but they cannot give blood to everyone. To whom can they *not* give blood?